

ARQ Electronics Student Projects Instructions & Process Outline

First, you email me your board artwork (Gerber files) and a bill of materials (BOM – a .csv or .xlsx file) listing each unique component used on your board. See the example BOM below to get an idea of what sort of information it should contain. Using the provided BOM template (“Student Project BOM Template.xlsx”) is not required but does help speed up the quoting process.

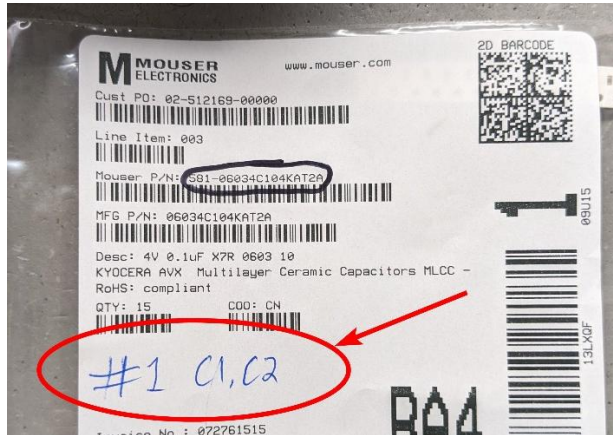
#	Quantity	Mfgr's Part #	Ref Designators	Description	TYPE	# of Leads
1	2	06034C104KAT2A	C1, C2	0.1µF 0603 cap	SMD	2
2	3	CRCW12064K02FKEA	R1, R4, R6	4.02 kOhm Res, 1206	SMD	2
3	2	705410040	J1, J2	CONN, HDR, MOLEX-SL, 6P, ST	TH	6
4	2	LTL-4233	LED_standby, LED_status	LED, GREEN, T1-3/4, DIFFUSED	TH	2
5	1	AD8629ARZ	U1	IC, ANA, OPAMP, Dual, Low Noise	SMD	8
		DO NOT POPULATE	R5, J3, LED_pwr			

- BOM
 - Note: a BOM lists the parts & quantities needed to make a *single* assembly. If you are having 3 boards assembled and each board needs 1 of component #5, then line #5 should list “1” as the quantity.
 - The columns highlighted blue are required. Information such as type (Surface Mount vs Through Hole) and number of leads are not required but *do help speed up the quoting process to get you a quote as soon as possible*.
 - Ensure each reference designator on the board artwork is listed on the BOM
 - If any part locations on the board are not to be populated, list them in the BOM as “DO NOT POPULATE” or “DNP”
- Artwork (Gerber files)
 - Ensure that any parts with polarity, numbered pins, or directionality are clearly marked to indicate their orientation.
 - Parts with numbered pins are typically marked by a dot next to pin 1, or for thru-hole, by having a square pad for pin 1
 - Polarity is typically marked with a dot or plus sign next to the positive terminal
 - Although most surface mount capacitors are not polarized, some are. Tantalum caps (typically yellow) are polarized
 - Handwritten notes on the component bags can also be used to clarify orientation (see below)

Second, you bring your kit, which includes both the bare PCB and all the components necessary to assemble your project, to ARQ.

- **NOTE: Our hours for student projects are Mon-Thur, 9 AM to 3:45 PM**
 - **The front doors are kept locked during business hours and the front desk isn't always attended. Therefore, let us know ahead of time when you would like to drop off or otherwise visit ARQ. Once you arrive, if the doors are locked, please call the receptionist at 979-696-5889 and tell her your name and let her know I am expecting you. This number is also posted on the front door.**

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- Each unique part (line in the BOM) should be in its own bag (the distributor bag, e.g. from Mouser, Digikey, is fine)
 - On each bag, you must write the # and reference designators corresponding to that part.
 - For example, based on the example BOM above, on the part bag for 06034C104KAT2A, you could write “#1” and “C1, C2” on the bag, as shown in the photo on the left.
 - Some distributors let you include notes to be printed directly on the label for each component, this is also an acceptable way to mark the number and reference designators.
- If there is any part orientation not marked on the board itself, you should clarify with a note on the bag such as “+ goes toward J2 connector” or “pin 1 is closest to corner” or “switch ON position faces towards board logo”
- Sort the bags in order based on their #s in the BOM
- If there is any conflict between the BOM and what is written on your bags, we will likely have to contact you for clarification, leading to delays in you receiving your project, so please carefully check everything beforehand

Third, a technician will estimate the time/cost to complete your project. I will produce a quotation for the project and send it to you, your professor, and/or your department business office.

- Which department you are in will determine to whom the quote should be sent for approval.
 - ARQ already has a working relationship with ETID, so not much is required on your end if you are in this department.
 - If you are in any other department, you will need to let me know to whom I should send the quote.

Fourth, once the quote is approved, I will receive a purchase order (PO) for the project from your department. Once a PO is received, we will begin assembly on your board (unless we are still waiting to receive any components for the assembly).

Our standard lead time is 10 business days after receiving a PO and all components, but there are several things that can affect this. For very densely populated boards, it may be necessary for ARQ to order a solder paste stencil, which increases lead time. Boards with BGA (ball grid array) components may also take longer than the standard lead time.

However, it is not uncommon for a project to be ready sooner. The best thing you can do to get your project back quickly is to follow the guidelines for your artwork, BOM, and kit. If other jobs ahead of yours are held up waiting on answers that don't come quickly and the tech sees your kit and documentation are in line with the above, they will be more likely to move to yours while they wait for clarifications on other jobs. Once they are on a job, they'll complete it, if there are no further roadblocks, before going back to a previous job. So, you can help give us the best chance at completing your project in the shortest amount of time by following the above guidelines.

Fifth, once your assembly is complete, I will email you and we will arrange a time for you to pick up your project. When you pick up your project, you will sign and date the PO from your department and an invoice from ARQ as proof of delivery.

This guide is a work in progress. Please do not hesitate to contact me with any further questions or clarifications. jkdillon@arqelectronics.com or fjdrummond@arqelectronics.com